

Churn prediction documentation

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→Dataset: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

The dataset predicts behavior to retain customers.

Content

Each row represents a customer, each column contains customer's attributes described on the column Metadata.

The data set includes information about:

- Customers who left within the last month – the column is called Churn
- Services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
- Customer account information – how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges
- Demographic info about customers – gender, age range, and if they have partners and dependents

Attribute	Description	Possible Values / Type
customerID	Unique identifier for each customer	String
gender	Customer's gender	Male, Female
SeniorCitizen	Indicates whether the customer is a senior citizen	0 = No, 1 = Yes

Attribute	Description	Possible Values / Type
Partner	Whether the customer has a partner	Yes, No
Dependents	Whether the customer has dependents	Yes, No
tenure	Number of months the customer has stayed with the company	Integer (Months)
PhoneService	Whether the customer has a phone service	Yes, No
MultipleLines	Whether the customer has multiple phone lines	Yes, No, No phone service
InternetService	Customer's internet service provider	DSL, Fiber optic, No
OnlineSecurity	Whether the customer has online security service	Yes, No, No internet service
OnlineBackup	Whether the customer has online backup service	Yes, No, No internet service
DeviceProtection	Whether the customer has device protection	Yes, No, No internet service
TechSupport	Whether the customer has technical support	Yes, No, No internet service
StreamingTV	Whether the customer subscribes to streaming TV	Yes, No, No internet service
StreamingMovies	Whether the customer subscribes to streaming movies	Yes, No, No internet service

Attribute	Description	Possible Values / Type
Contract	Customer's contract type	Month-to-month, One year, Two year
PaperlessBilling	Whether paperless billing is enabled	Yes, No
PaymentMethod	Customer's payment method	Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic)
MonthlyCharges	Monthly amount charged to the customer	Float
TotalCharges	Total amount charged to the customer	Float
Churn (<i>Target Variable</i>)	Indicates whether the customer left the company	Yes, No

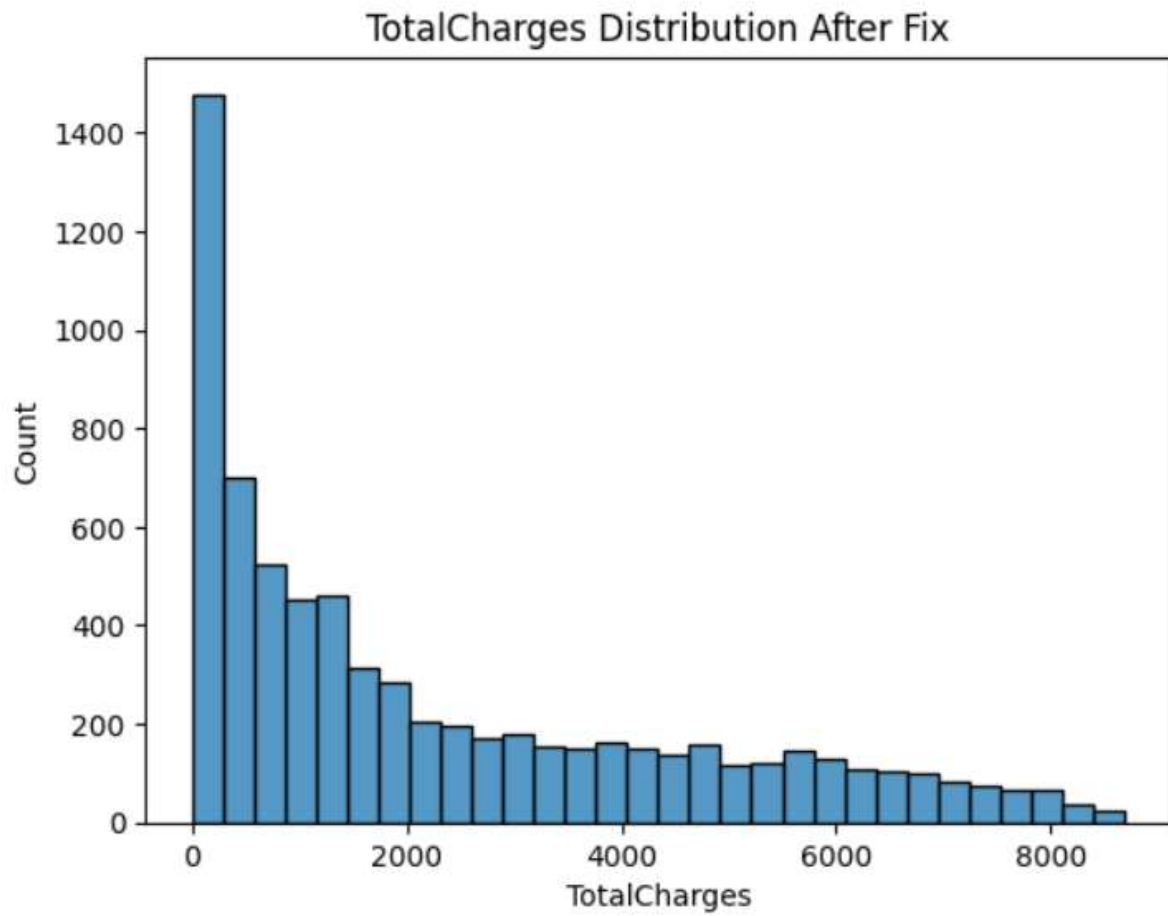
→Milestones :

Milestone 1: In this milestone we focused on preprocessing the data :

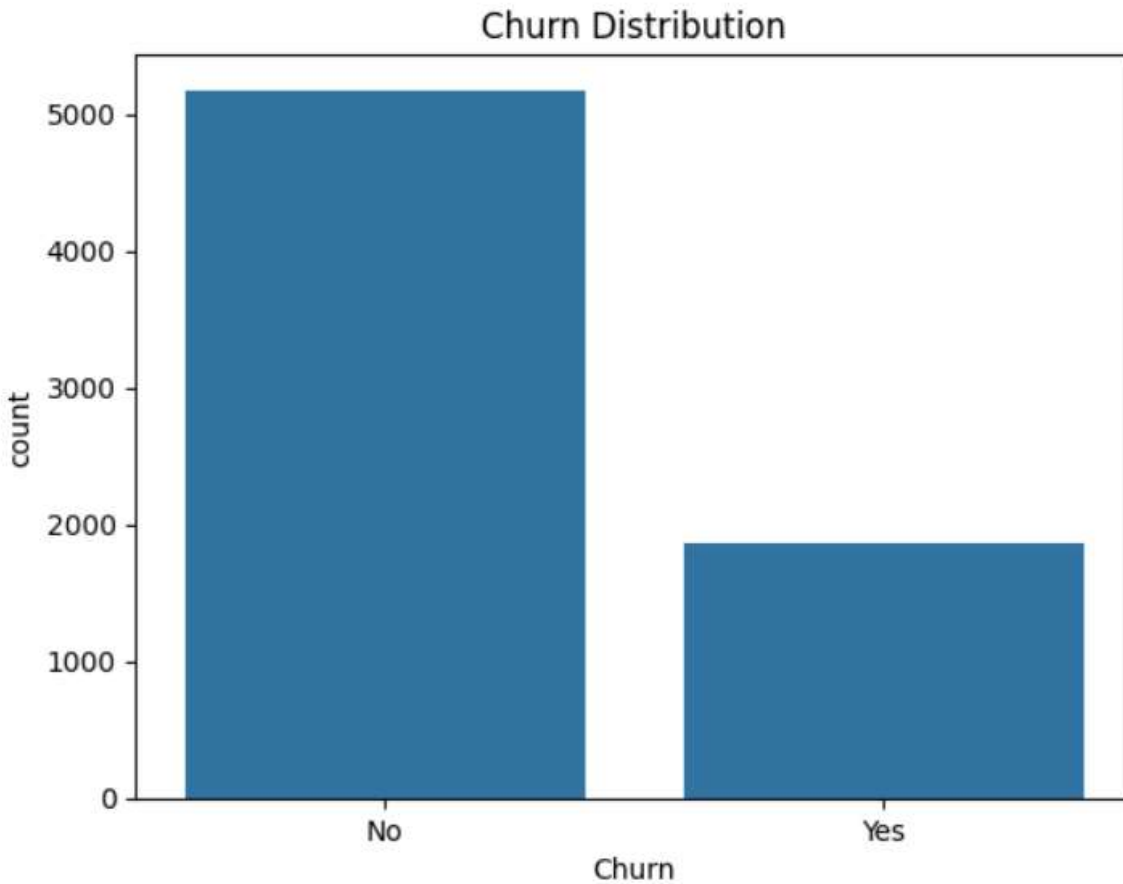
1st : Data loading & initial overview: We loaded the data from Kaggle, and we found that the data has 21 columns: 1 of type float , 2 of type int and 18 of type object

2nd : analyzing the data: we didn't find any null values and no duplicates, but we found garbage values

We fixed the “total charges “column because it has Nan values, so we first convert it to numerical values and then fill the blanks with 0.



Regarding the Imbalance classes:



The dataset is imbalanced, meaning one class ("No") has significantly more samples than the other ("Yes").

Although the imbalance is not extreme, it is important because:

- A machine learning model could become biased toward predicting "No".

3rd feature extraction using sql:

→ We found 73.46% stayed and 26.54% churned

Query Results

Contract Type	Total Customers	Churned Customers	Churn Rate
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two year	1,695	48	2.83%

- **Month-to-month contracts have the highest churn rate (42.71%). Nearly 4 out of every 10 customers with a monthly contract left the company. This suggests that customers without a long-term commitment are much more likely to switch providers.**

- **Customers with a one-year contract have a much lower churn rate of 11.27%, indicating that longer commitments improve customer retention.**

- **Two-year contracts have the lowest churn rate (2.83%), meaning almost all customers with long-term contracts remain with the company.**

→The results reveal a strong negative relationship between contract length and customer churn. As the contract duration increases, the likelihood of a customer leaving the company decreases significantly.

This suggests that contract type is one of the strongest predictors of customer churn in the dataset.

Query Results

Internet Service	Total Customers	Churned Customers	Churn Rate
Fiber optic	3,096	1,297	41.89%
DSL	2,421	459	18.96%
No Internet	1,526	113	7.40%

- **Fiber optic customers have the highest churn rate (41.89%). Nearly 42% of customers using fiber optic internet left the company. This is more than twice the churn rate of DSL customers and almost six times that of customers without internet service.**
- **DSL customers have a moderate churn rate (18.96%). While some customers leave, they are considerably more likely to stay compared to fiber optic users.**
- **Customers with no internet service have the lowest churn rate (7.40%), indicating that they are the most loyal group in the dataset.**

The analysis suggests that internet service type is strongly associated with customer churn. In particular, Fiber optic customers are the most likely to leave the company.

→Query Results

Churn Status	Avg. Monthly Charges	Avg. Total Charges	Avg. Tenure (Months)
No	61.27	2549.91	37.57
Yes	74.44	1531.80	17.98

Interpretation

1. Monthly Charges

- Customers who churned paid an average of 74.44 per month.
- Customers who stayed paid an average of 61.27 per month.

Observation:

Customers with higher monthly charges are more likely to leave the company.

2. Total Charges

- Customers who stayed paid an average total of 2,549.91.
- Customers who churned paid only 1,531.80 on average.

Observation:

Although churned customers pay more each month, they have lower total charges because they remain with the company for a shorter period.

3. Customer Tenure

- Customers who stayed have an average tenure of 37.57 months.
- Customers who churned stayed for only 17.98 months on average.

Observation:

Customers who leave tend to do so relatively early in their relationship with the company.

Business Insight

The analysis reveals three important patterns:

- Customers with higher monthly bills are more likely to churn.
- Customers who churn have much shorter tenures, indicating that the first few months are critical for customer retention.
- Since they leave earlier, churned customers accumulate lower total charges despite paying more per month.

→Query Results

Customer ID	Tenure (Months)	Monthly Charges	Contract	Churn
3292-PBZEJ	11	111.40	Month-to-month	No
9851-KIELU	10	110.10	Month-to-month	Yes
3992-YWPKO	6	109.90	Month-to-month	Yes
5760-IFJOZ	3	107.95	Month-to-month	No
2081-VEYEH	3	107.95	Month-to-month	No
1400-MMYXY	3	105.90	Month-to-month	Yes
3389-YGYAI	8	105.50	Month-to-month	Yes
5052-PNLOS	3	105.35	Month-to-month	Yes
6734-GMPVK	5	105.30	Month-to-month	No
4587-VVTOX	6	105.30	Month-to-month	Yes

Interpretation

Several common characteristics can be observed among these customers:

- All customers have month-to-month contracts, the contract type with the highest churn rate in the dataset.
- Their monthly charges exceed \$105, which is significantly higher than the overall average monthly charge.

- Most have been customers for less than one year, indicating short tenure.
- Among these ten customers, 6 have already churned, while 4 remain active, suggesting that the active customers share many of the same risk factors and may be likely to churn if no retention efforts are made.

Business Insight

This analysis highlights a customer segment that deserves special attention. Customers who are new, pay high monthly fees, and are on month-to-month contracts appear to have a much higher likelihood of leaving the company.

→Query 6: Churn Rate by Payment Method

Business Insights

The analysis shows that payment method has a significant impact on customer churn.

Customers who pay using Electronic Check have the highest churn rate (45.29%), which is nearly three times higher than customers using automatic payment methods. In contrast, customers who pay through Credit Card (Automatic) and Bank Transfer (Automatic) have the lowest churn rates (15.24% and 16.71%, respectively).

This suggests that customers enrolled in automatic payment methods are generally more loyal and less likely to discontinue the service.

→Query 7: Senior Citizens Churn Rate

Business Insights

Senior citizens have a substantially higher churn rate (41.68%) compared to non-senior customers (23.61%).

→Query 8: Number of Services per Customer

Business Insights

A new feature called `total_services` was created to measure customer engagement by counting the number of subscribed services.

Customers with the maximum number of subscribed services (8 services) are all active customers (Churn = No) in the sample shown. This suggests that customers who subscribe to multiple services tend to be more engaged with the company and are less likely to churn.

→A new feature called `total_services` was created by counting the number of subscribed services for each customer, including phone service, multiple lines, online security, online backup, device protection, technical support, streaming TV, and streaming movies.

This feature represents the customer's level of engagement with the company's services. Customers with a higher number of subscribed services are generally more invested in the company's offerings and are expected to have a lower likelihood of churning.

The new feature was successfully merged into the main dataset, allowing it to be used in subsequent exploratory analysis and predictive modeling.

4th .Exploratory Data Analysis

→We dropped 'customerID' column so the final columns

Demographics

'gender', 'SeniorCitizen', 'Partner', 'Dependents',

Services

'PhoneService', 'MultipleLines', 'InternetService',

'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',

'TechSupport', 'StreamingTV', 'StreamingMovies',

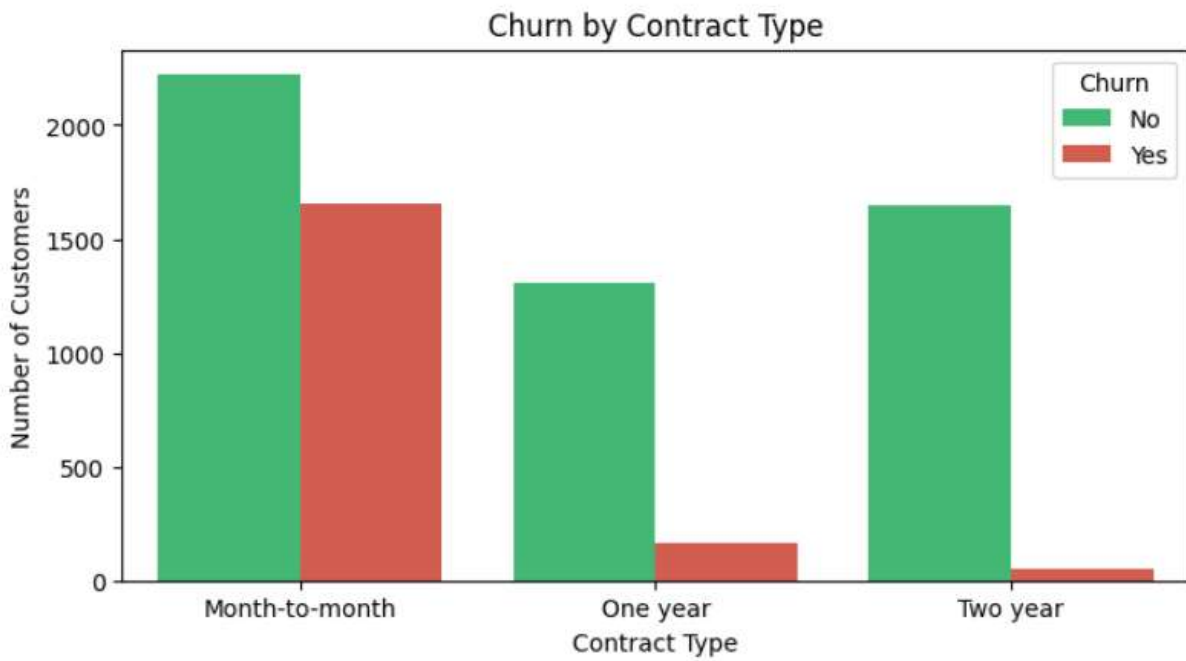
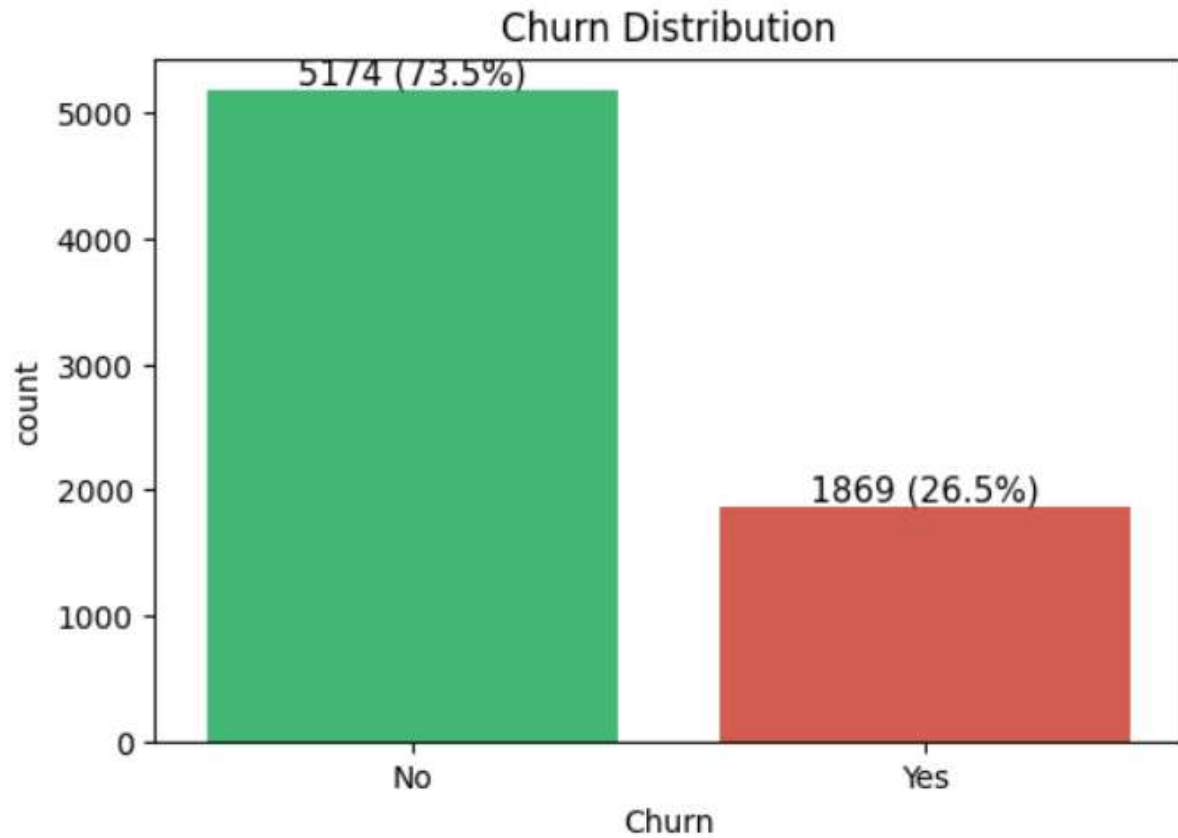
Billing

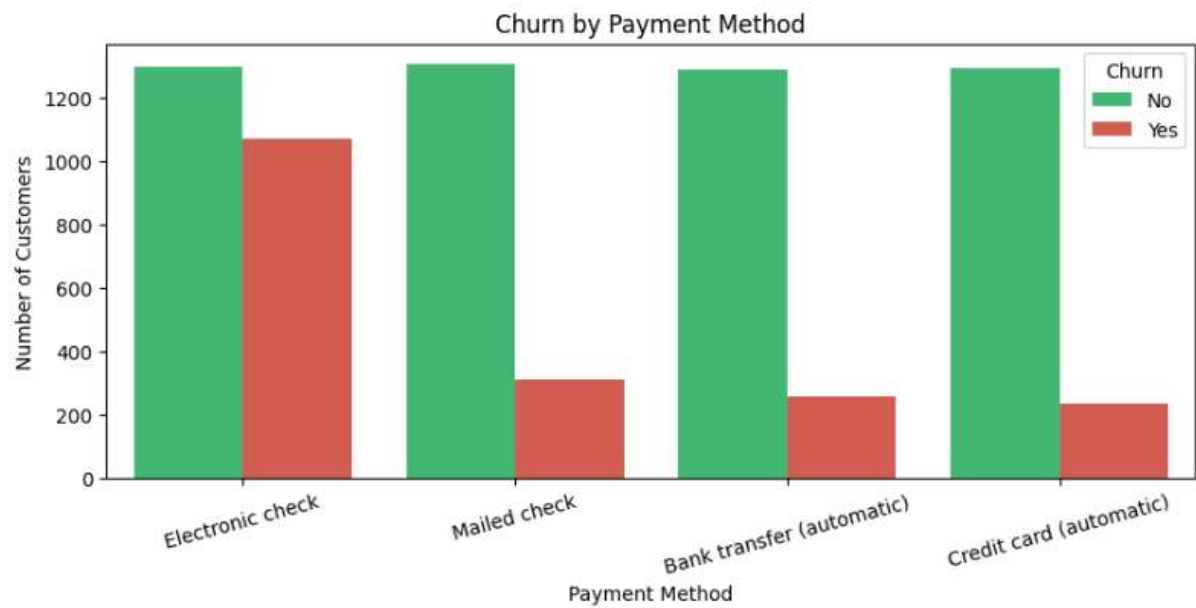
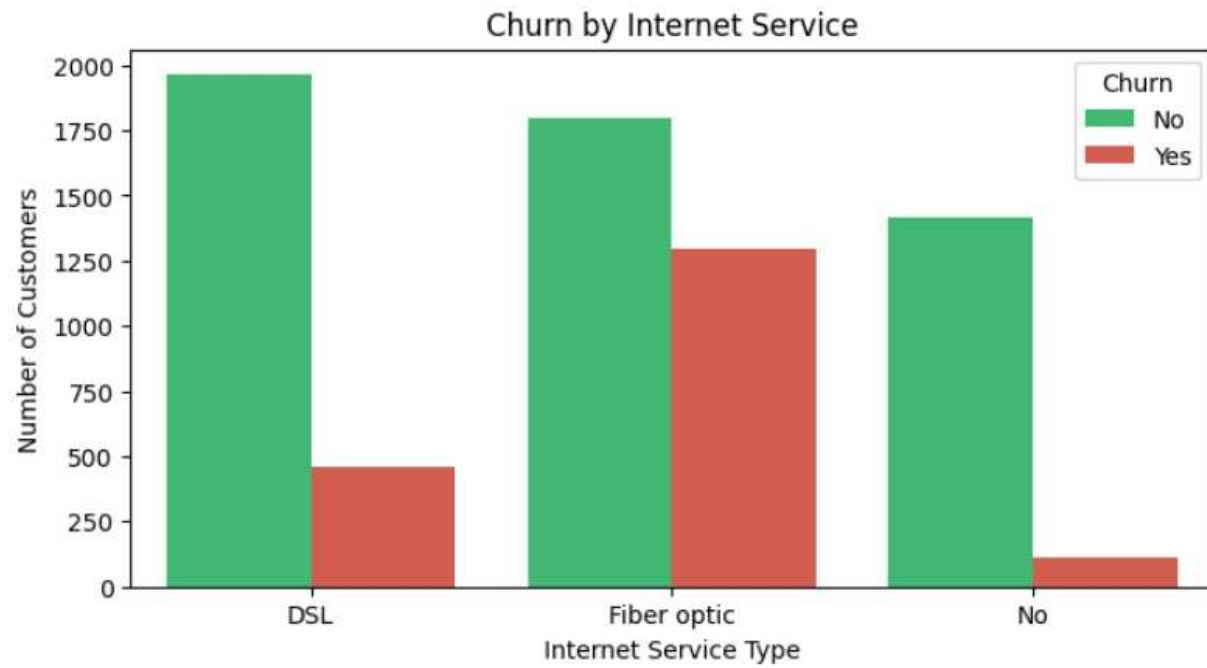
'Contract', 'PaperlessBilling', 'PaymentMethod'

→we found that 'gender ' is the most weak column

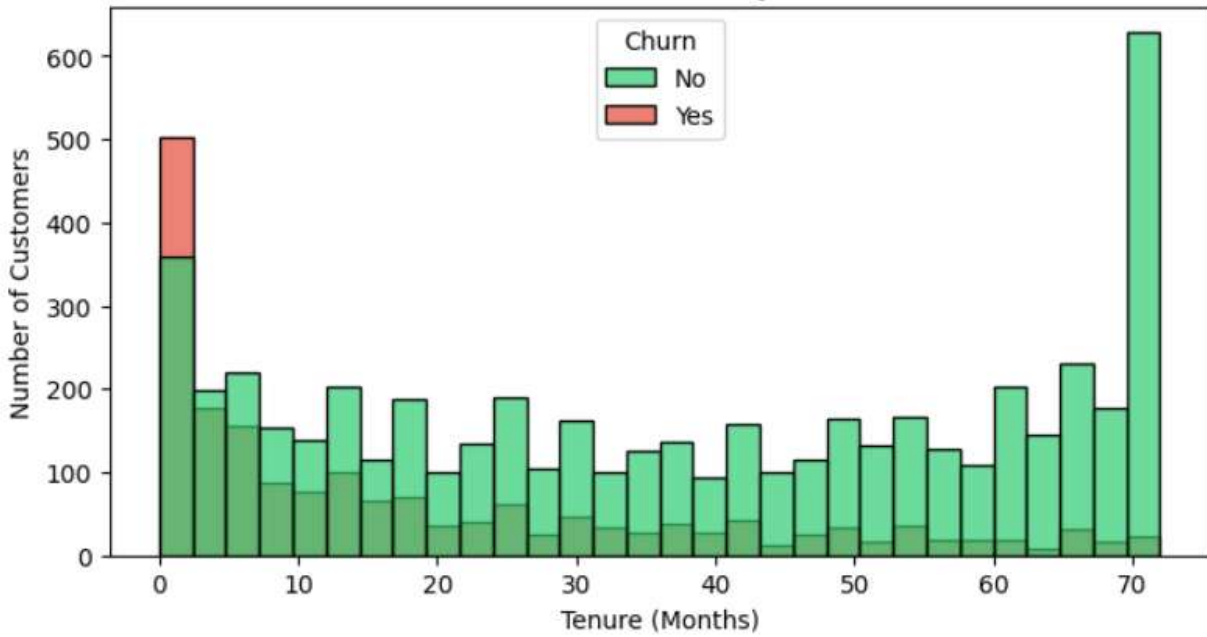
→we found that StreamingTV,StreamingMovies,MultipleLines,PhoneService are weak predictors

→Data integration and visualization:

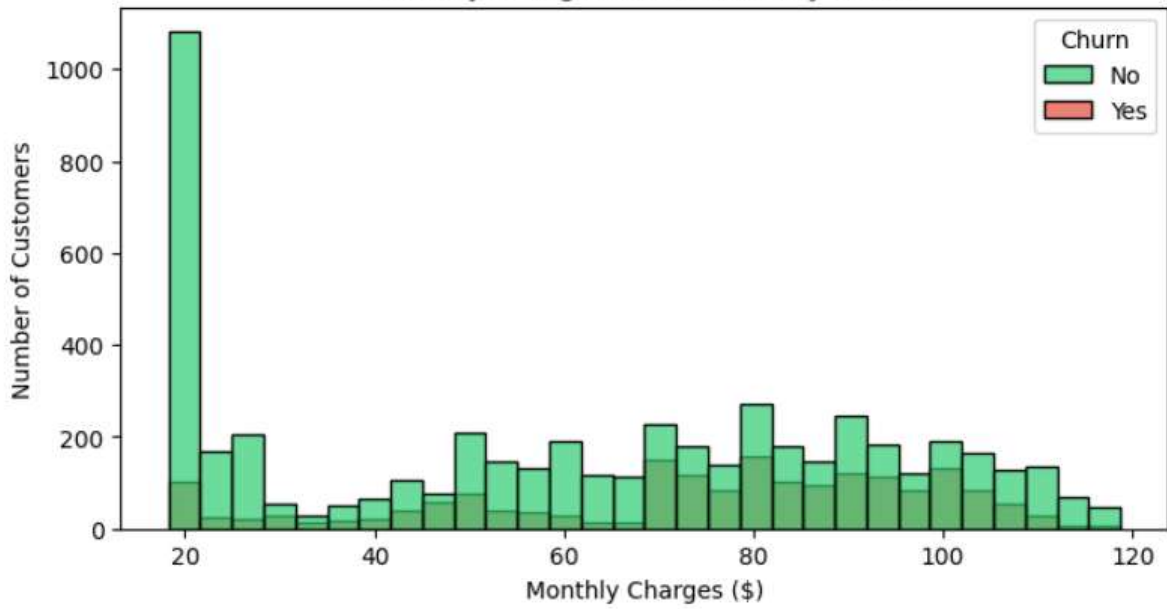




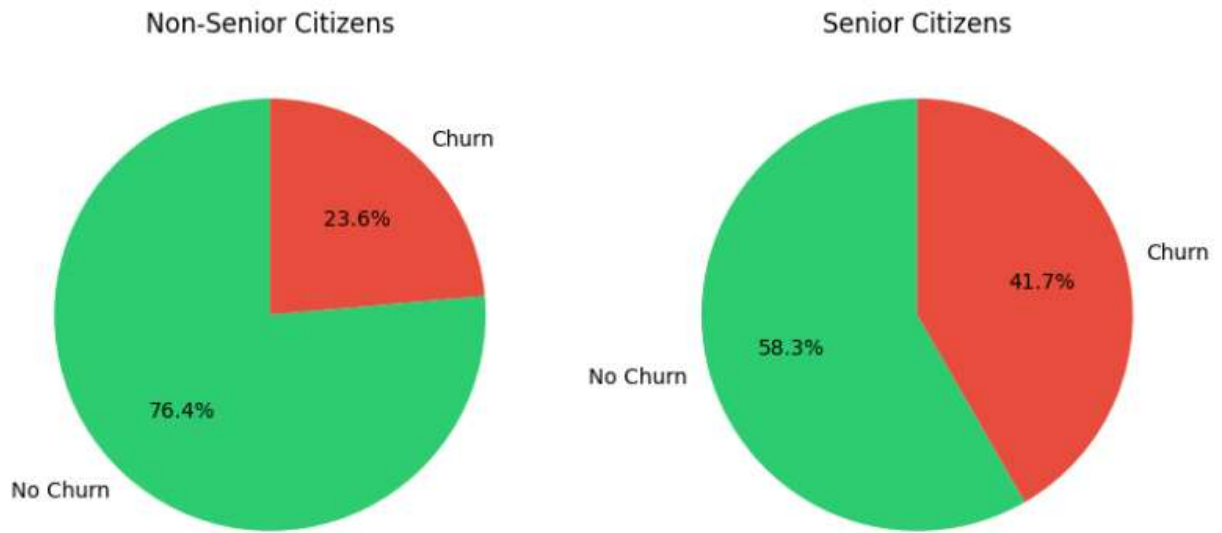
Tenure Distribution by Churn



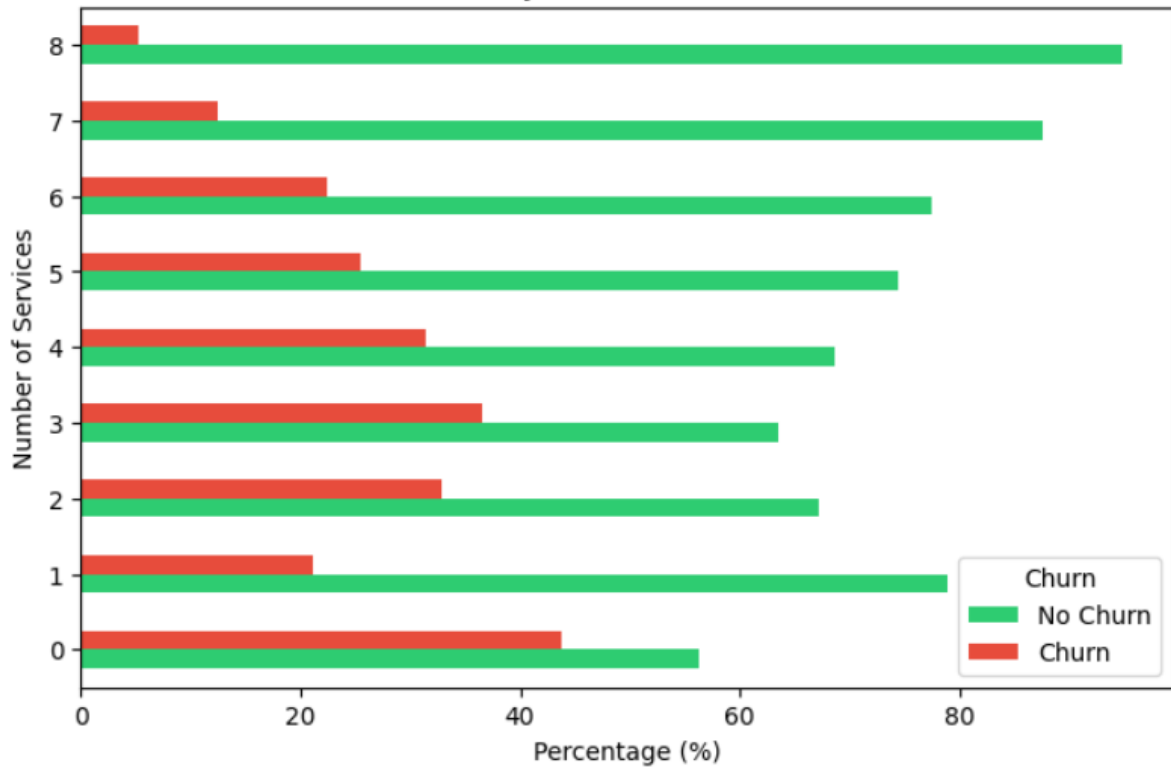
Monthly Charges Distribution by Churn

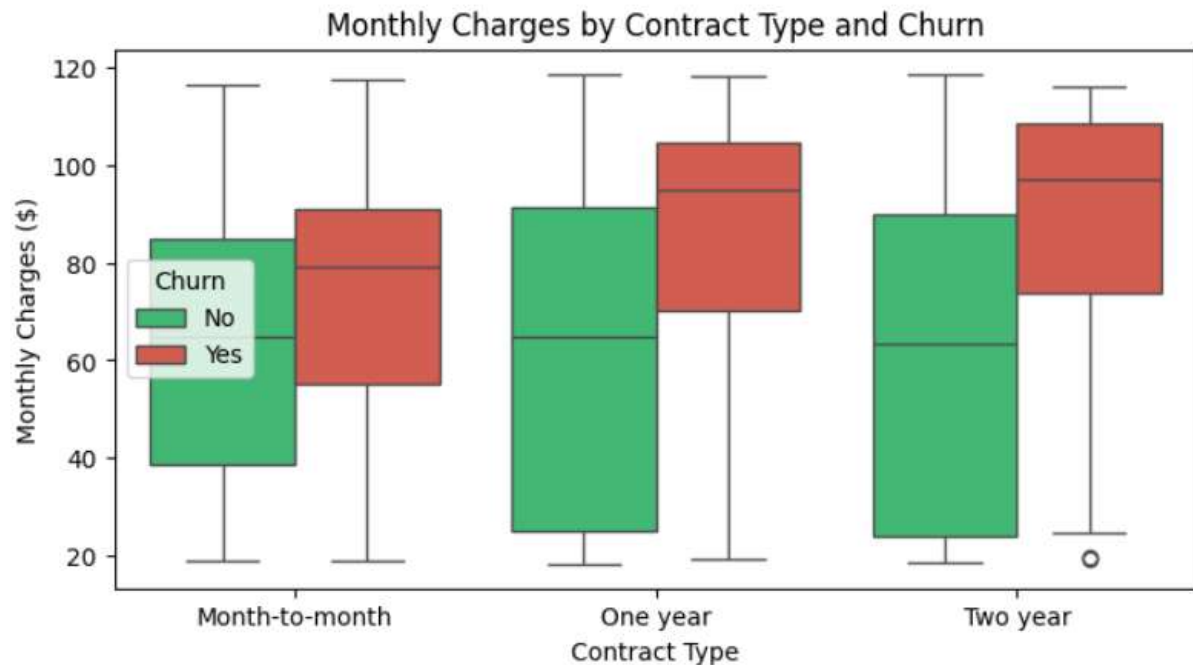


Churn Rate: Senior vs Non-Senior Citizens



Churn Rate by Total Services Subscribed





→Conclusions for each chart: **Viz 1 — Churn Distribution** 5174 customers stayed (73.5%) vs 1869 churned (26.5%). The dataset is imbalanced — this is why SMOTE is mandatory later.

Viz 2 — Churn by Contract Type Month-to-month customers churn massively compared to one-year and two-year contracts. Contract type is the single strongest predictor in the entire dataset. The company should push customers toward longer contracts.

Viz 3 — Churn by Internet Service Fiber optic customers churn the most despite — or because of — paying the highest prices. DSL customers are more stable. Customers with no internet barely churn at all.

Viz 4 — Churn by Payment Method Electronic check users churn at nearly 45% — the highest of any payment method. Automatic payment users (bank transfer and credit card) are the most loyal. Manual payment = less commitment.

Viz 5 — Tenure Distribution by Churn Churners are heavily concentrated in the first 12 months. The longer a customer stays, the less likely they are to leave. If you keep a customer past month 12 they are much safer.

Viz 6 — Monthly Charges Distribution by Churn Customers paying between 60-100/month churn the most. Low-charge customers (20-40) are very stable. High monthly cost combined with a short-term contract is the danger zone.

Viz 7 — Senior Citizen Pie Charts Senior citizens have a 41.7% churn rate vs only 23.6% for non-seniors. Even though seniors are a small group (16% of customers), they are a high-risk segment that needs targeted retention.

Viz 8 — Total Services Horizontal Bar Customers with 0–2 services churn heavily. Customers with 5+ services barely churn at all. This validates the total_services feature we engineered in SQL — it is a strong predictor.

Viz 9 — Boxplot Monthly Charges by Contract Month-to-month churners have the highest and widest spread of monthly charges. Two-year customers have low and consistent charges regardless of churn status. High charges + no long-term commitment = highest churn risk.

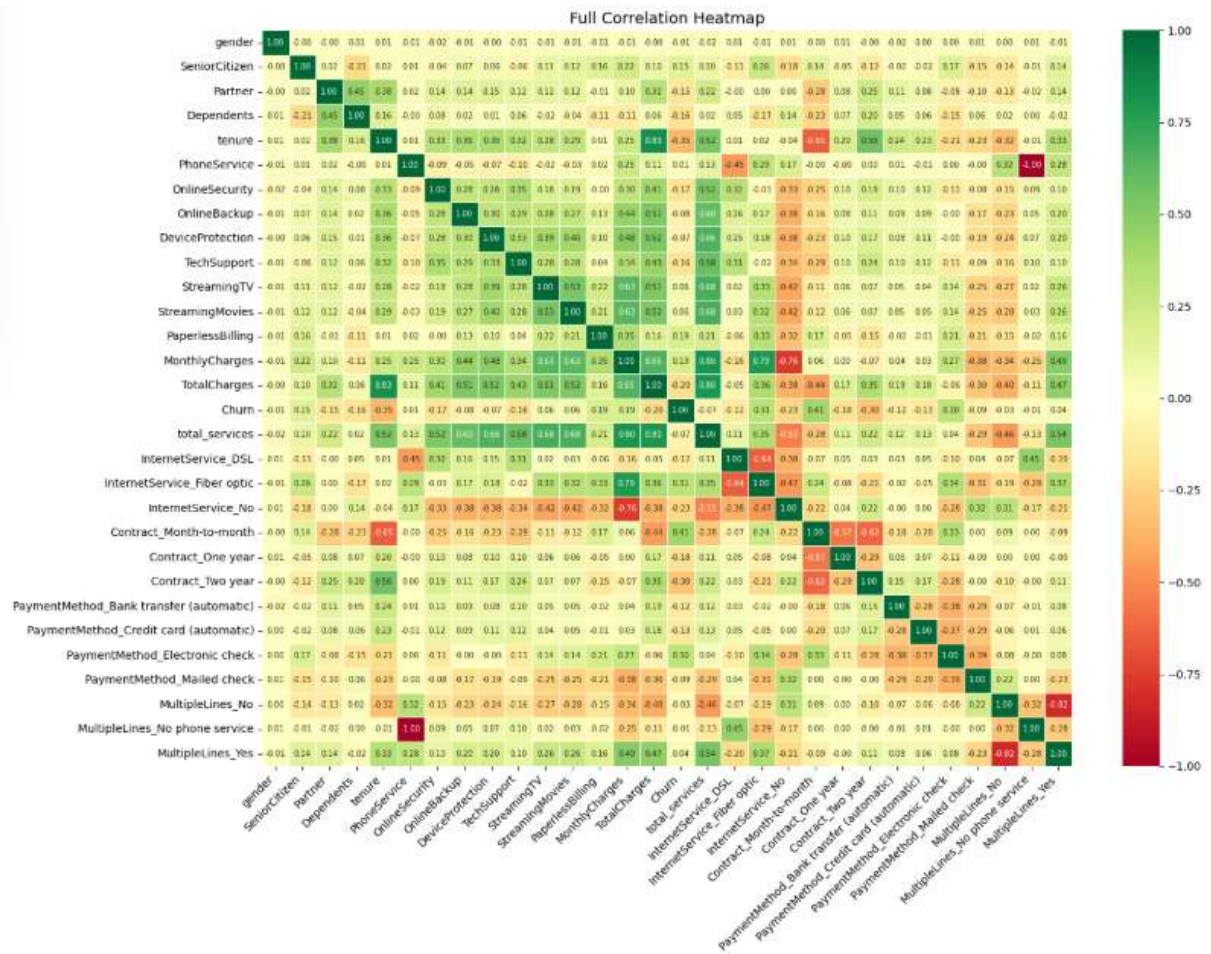
5th : feature engineering & encoding

1. Encoded 'gender' [female:0 , male: 1]
2. Encoded 'churn' [yes: 1, No : 0]
3. Encoded the binary columns as ['Partner', 'Dependents', 'PhoneService', 'PaperlessBilling', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies'] as following:

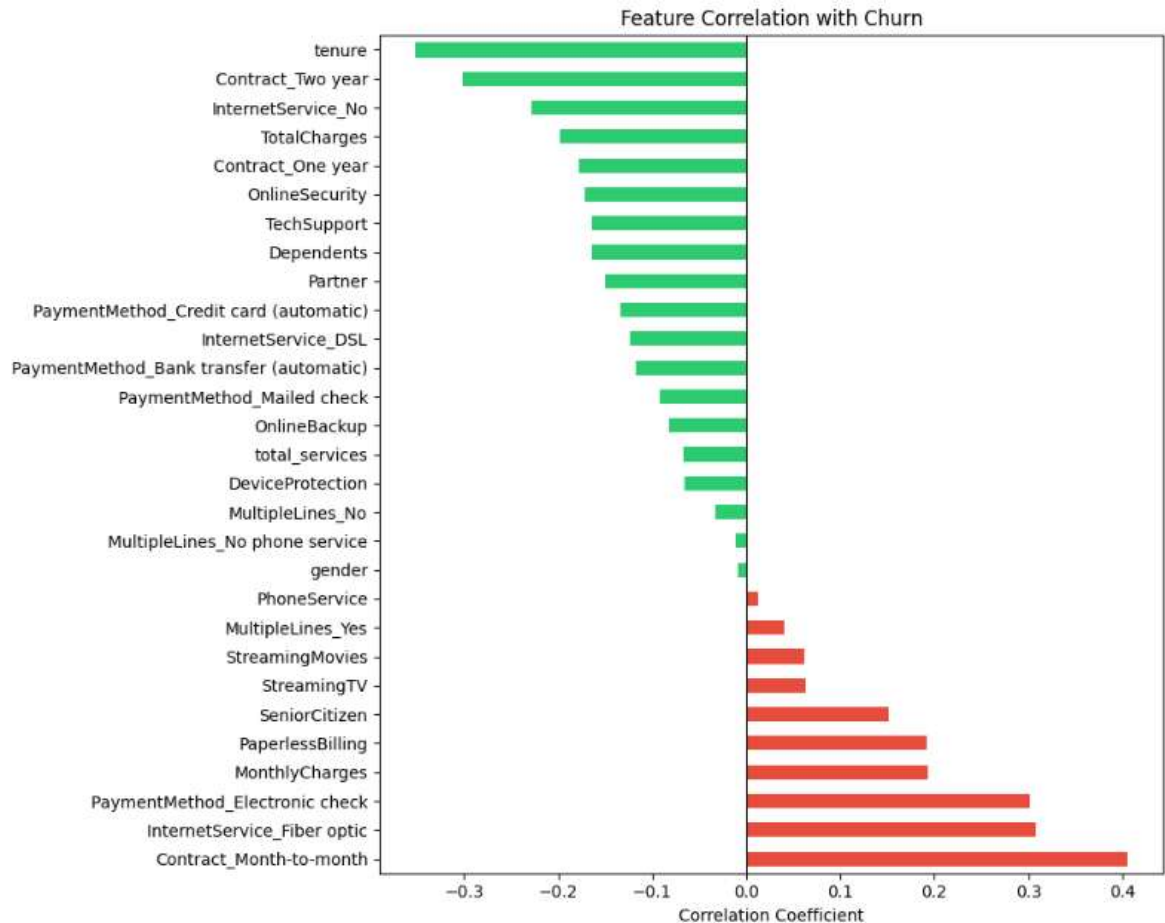
```
{ 'Yes': 1, 'No': 0, 'No internet service': 0  
'No phone service': 0 }
```

4. Applied one – hot encoding on ['InternetService', 'Contract', 'PaymentMethod', 'MultipleLines']

6th : correlation Analysis :



Correlation with target ' churn':



We'll drop the weak features that are smaller than the threshold 0.05:

['MultipleLines_Yes', 'PhoneService', 'gender', 'MultipleLines_No phone service', 'MultipleLines_No']

7th . Handling class imbalance:

We used SMOTE (Synthetic Minority Over-sampling Technique) to handle imbalanced datasets by creating synthetic (artificial) samples for the minority class instead of simply duplicating existing ones.

Before SMOTE:

Churn

0 4139

1 1495

After SMOTE:

Churn

0 4139

1 4139

We used feature scaler (MinMaxScaler() as a scaler) to normalize numerical features so they all have the same scale, typically between 0 and 1.

8th Export Artifacts:

To ensure reproducibility and simplify the next phase of the project, all preprocessing artifacts were exported to an artifacts directory. The balanced training set (after SMOTE), the untouched test set, the fitted MinMaxScaler, and the ordered list of feature columns were saved. These artifacts allow the machine learning model and future prediction API to use the exact same preprocessing pipeline, ensuring consistent and reliable predictions without repeating the preprocessing steps.

Columns:

[

"SeniorCitizen",

"Partner",

"Dependents",

"tenure",

"OnlineSecurity",

"OnlineBackup",

"DeviceProtection",

"TechSupport",

"StreamingTV",

"StreamingMovies",

"PaperlessBilling",

"MonthlyCharges",

```
"TotalCharges",
"total_services",
"InternetService_DSL",
"InternetService_Fiber optic",
"InternetService_No",
"Contract_Month-to-month",
"Contract_One year",
"Contract_Two year",
"PaymentMethod_Bank transfer (automatic)",
"PaymentMethod_Credit card (automatic)",
"PaymentMethod_Electronic check",
"PaymentMethod_Mailed check"
]
```

Milestone 2:

1st: Loading needed libraries & artifacts from milestone 1

2nd : Train pre-trained models (XGBoost & Random Forest): Random Forest and XGBoost were selected because they are powerful ensemble algorithms well suited for binary classification problems on structured data. They can model complex, non-linear relationships between customer attributes and churn while handling mixed feature types effectively. Random Forest provides a robust baseline with good interpretability through feature importance, whereas XGBoost often delivers superior predictive performance by sequentially improving weak learners. Comparing both models helps identify the approach that best balances predictive accuracy and the ability to detect customers at risk of churning.

3rd : Inference script: we implemented a reusable function that takes a raw customer feature row and returns a churn prediction + probability. X_row: DataFrame with columns matching feature_columns exactly. Returns dict with prediction (0/1) and churn probability per row.

Example:

Customer 0: predicted=0 prob=0.0028 actual=0

Customer 1: predicted=1 prob=0.9467 actual=0

Customer 2: predicted=0 prob=0.2015 actual=0

4th: Evaluation report:

	Accuracy	Precision	Recall	F1	ROC-AUC
Model					
XGBoost	0.7679	0.5501	0.6898	0.6121	0.8272
Random Forest	0.7601	0.5350	0.7353	0.6194	0.8395

```

=== XGBoost Classification Report ===
              precision    recall  f1-score   support

   No Churn      0.88      0.80      0.83     1035
    Churn      0.55      0.69      0.61      374

 accuracy              0.77     1409
 macro avg      0.71      0.74      0.72     1409
 weighted avg   0.79      0.77      0.78     1409

=== Random Forest Classification Report ===
              precision    recall  f1-score   support

   No Churn      0.89      0.77      0.82     1035
    Churn      0.54      0.74      0.62      374

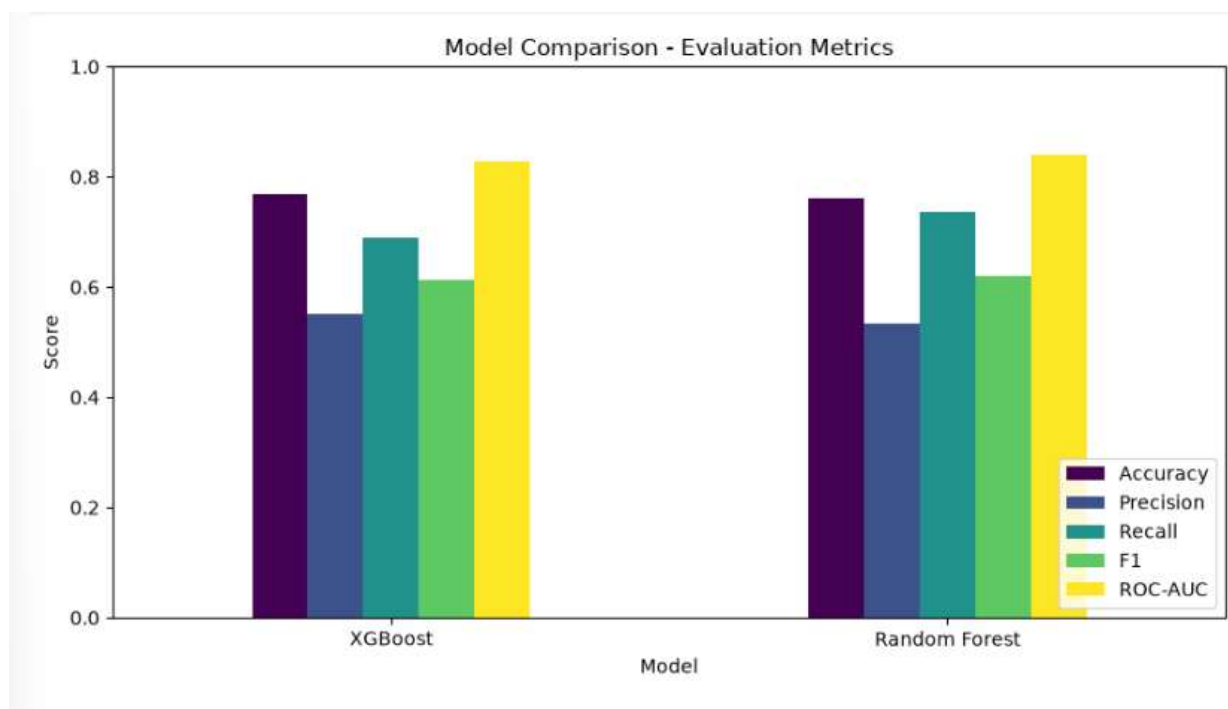
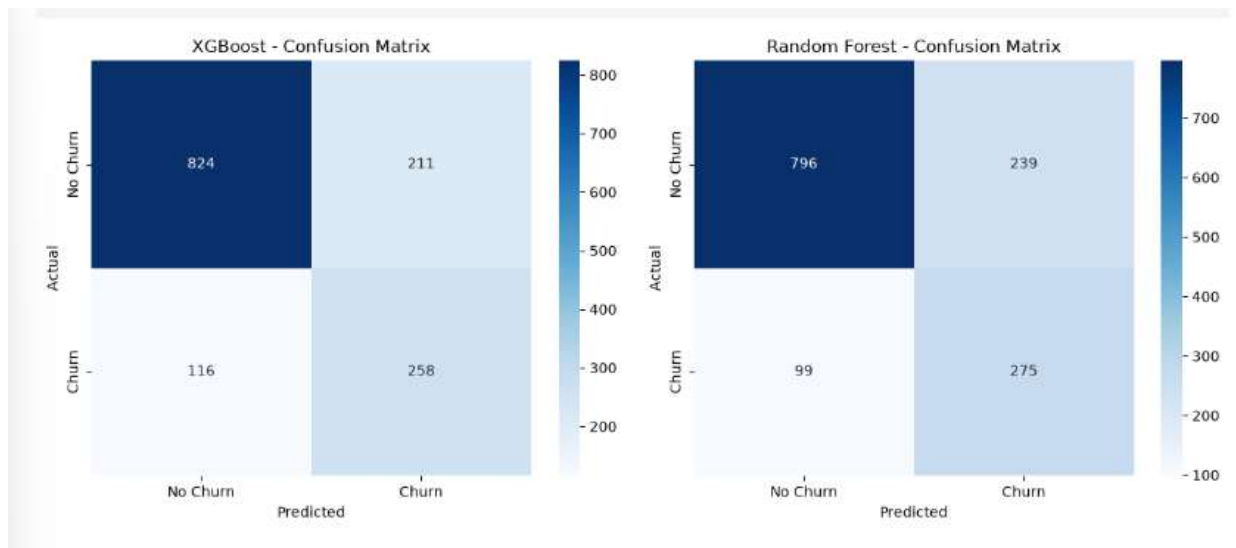
 accuracy              0.76     1409
 macro avg      0.71      0.75      0.72     1409
 weighted avg   0.80      0.76      0.77     1409

```

From the report above , we conclude that both models achieved similar overall performance, with accuracies close to 76–77%.

- XGBoost achieved the highest Accuracy and Precision, making it slightly better at avoiding false churn predictions.
- Random Forest achieved higher Recall, F1-score, and ROC-AUC, meaning it identified more churning customers and showed better overall classification performance.

5th : Confusion Matrices & Model Comparison Chart



6th : Logistic regression

A Logistic Regression model was trained as a baseline classifier and evaluated using the same test set as the ensemble models. Its performance metrics, along with the training time, were added to a comparison table containing XGBoost and Random Forest. The models were ranked based on ROC-AUC, allowing an objective comparison of their ability to distinguish between customers who churn and those who remain. Using Logistic Regression as a benchmark helps assess whether the increased complexity of ensemble methods results in meaningful performance

improvements.

	Accuracy	Precision	Recall	F1	ROC-AUC	TrainTime(s)
Model						
Random Forest	0.7601	0.5350	0.7353	0.6194	0.8395	NaN
Logistic Regression	0.7516	0.5227	0.7380	0.6120	0.8289	0.058
XGBoost	0.7679	0.5501	0.6898	0.6121	0.8272	NaN

	Accuracy	Precision	Recall	F1	ROC-AUC
Model					
XGBoost	0.7679	0.5501	0.6898	0.6121	0.8272
Random Forest	0.7601	0.5350	0.7353	0.6194	0.8395

Best model by ROC-AUC: Random Forest

Milestone 3:

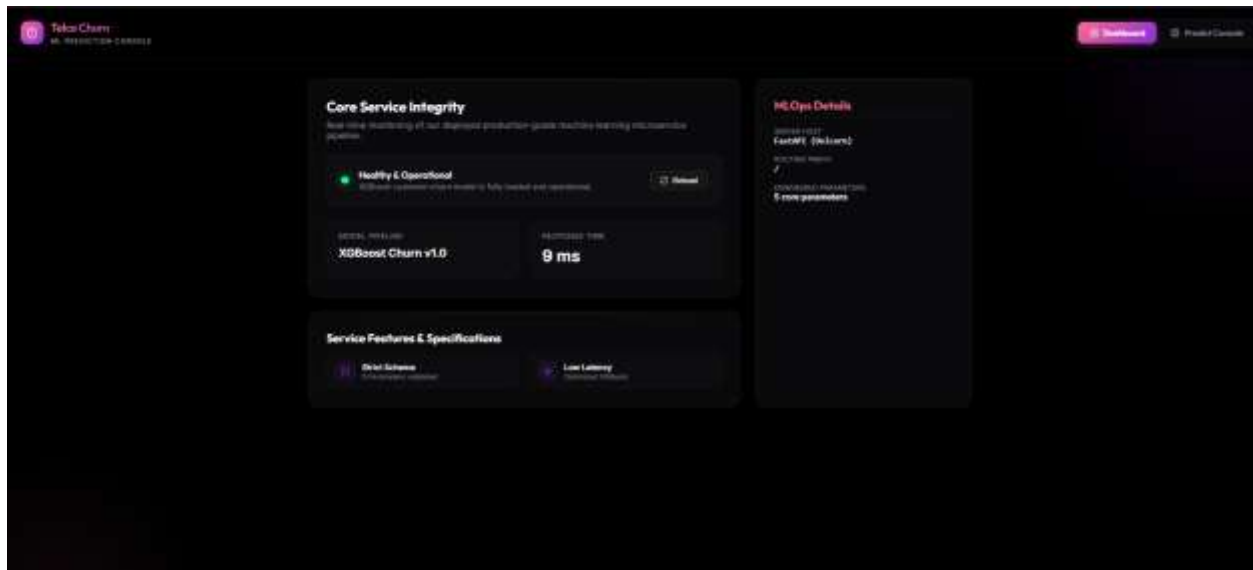
The system includes:

- FastAPI REST API
- Dockerized deployment
- Apache Airflow orchestration
- Prediction logging
- Monitoring dashboard using Power BI

Technologies Used

- Python
- FastAPI
- XGBoost

- Docker
- Docker Compose
- Apache Airflow
- Power BI
- Pandas



The screenshot displays the 'Telco Churn ML Prediction Console' dashboard with a form for predicting churn risk. The form is divided into two main sections: 'Monthly Charges (€)' and 'Fixed Charges (€)'. The 'Monthly Charges (€)' section includes fields for 'Monthly Charges (€)' (75.00), 'Contract Type' (One year), 'Service Level' (Fiber optic), and 'Payment Method' (Electronic check). The 'Fixed Charges (€)' section includes fields for 'Fixed Charges (€)' (945.00), 'Service Level' (Fiber optic), and 'Payment Method' (Electronic check). Below these sections is a 'DEMOGRAPHICS & ADD-ON SERVICES' section with a grid of dropdown menus for various services. A large circular progress indicator on the right shows '23% RISK OF CHURN' and 'LOW CHURN RISK'. At the bottom, there is a button labeled 'Analyze Churn Risk'.

The dashboard above shows a real – time monitoring of our deployed production-grade machine learning microservice pipeline , it shows that machine learning model has been successfully deployed and is ready to receive prediction requests.

There are 2 pages:

→ **Dashboard:** Displays the system status and performance.

→ **Predict Console:** Allows users to enter customer information and receive a churn prediction.

1st Dashboard:

There is a response time that shows the time Api takes to process a prediction request , the workflow works in the following

User sends customer information



FastAPI receives request



Preprocess input



Load XGBoost model



Predict churn



Return prediction

In our project, FastAPI acts as the bridge between the user interface and the machine learning model.

Instead of running the XGBoost model only inside a Python notebook, FastAPI exposes it as a REST API so that any application (web dashboard, mobile app, or another service) can send customer information and receive a churn prediction.

The main responsibilities of FastAPI in our project are:

- Receive prediction requests from the user.
- Validate the input data before it reaches the model.
- Apply the required preprocessing steps.
- Pass the processed data to the XGBoost model.

- **Return the prediction as a JSON response.**
- It uses **Pydantic** models to define the expected structure of incoming requests.

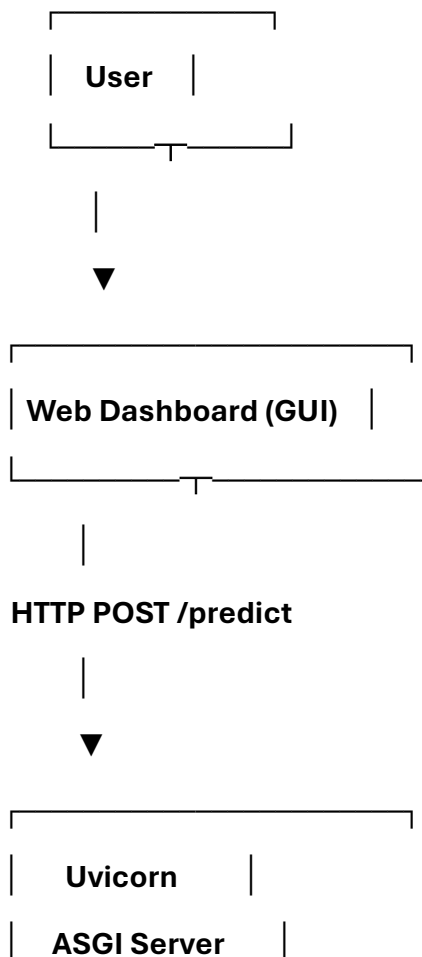
What is Uvicorn?

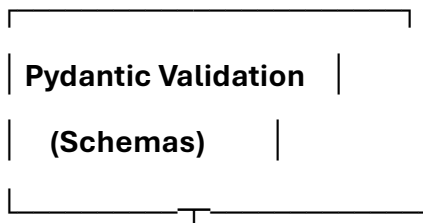
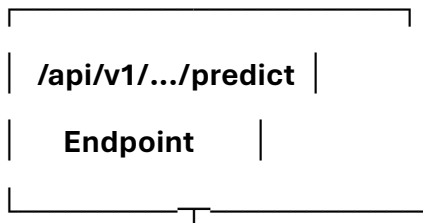
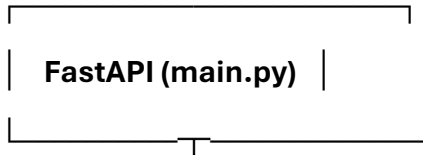
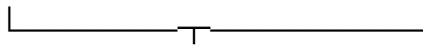
Uvicorn is a lightweight ASGI server that runs the FastAPI application and makes it accessible over the web.

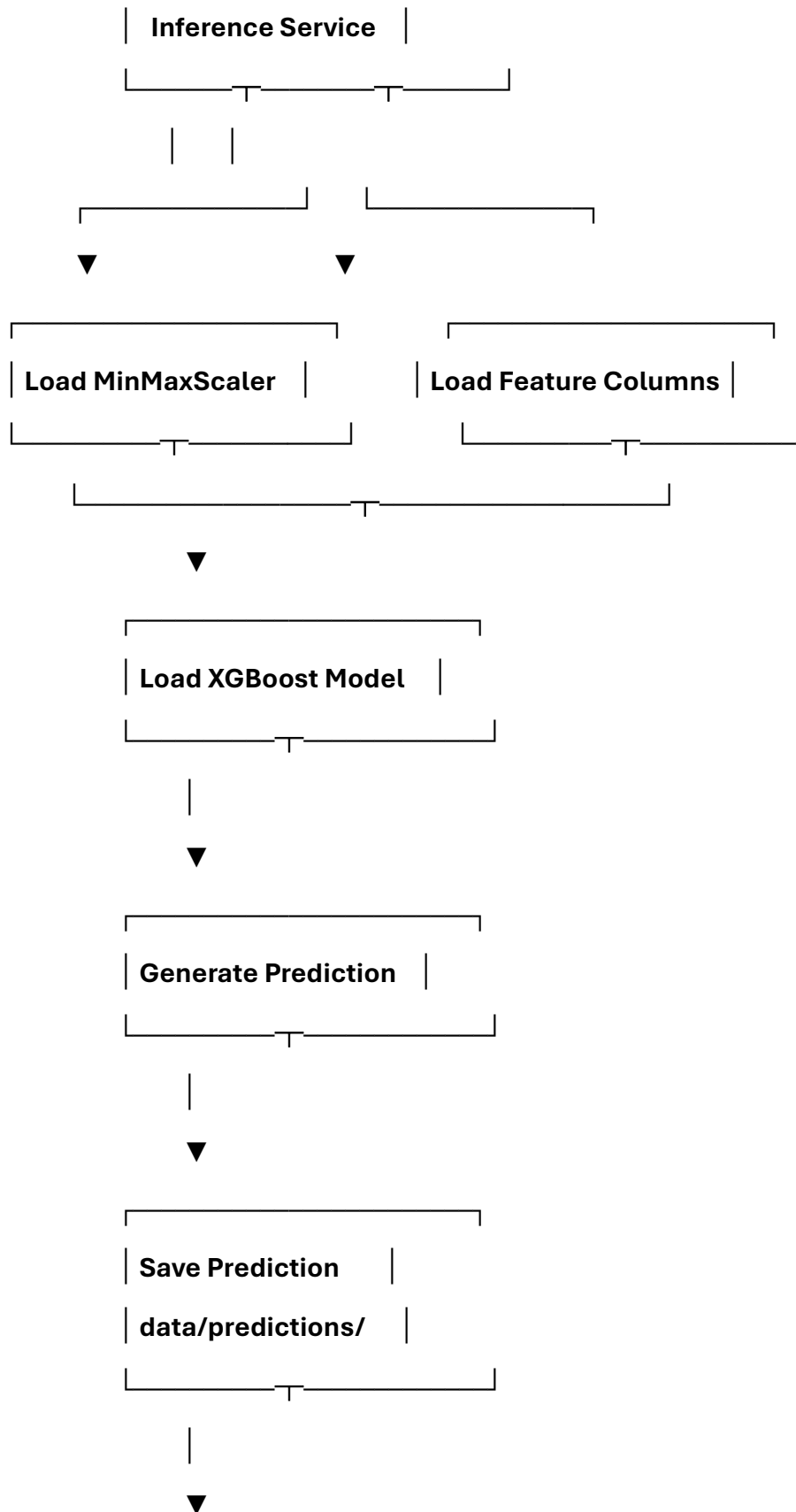
Its main responsibilities are:

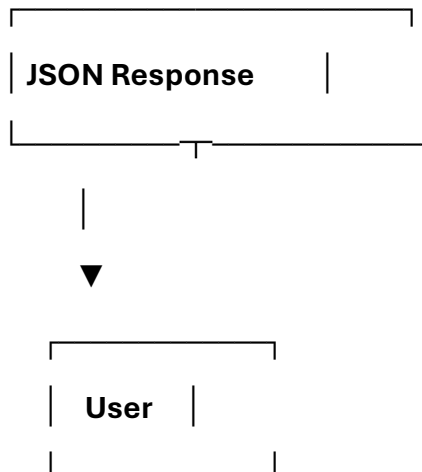
- **Starting the FastAPI application.**
- **Listening for incoming HTTP requests.**
- **Forwarding requests to FastAPI.**
- **Returning responses back to the client.**

The request flow:





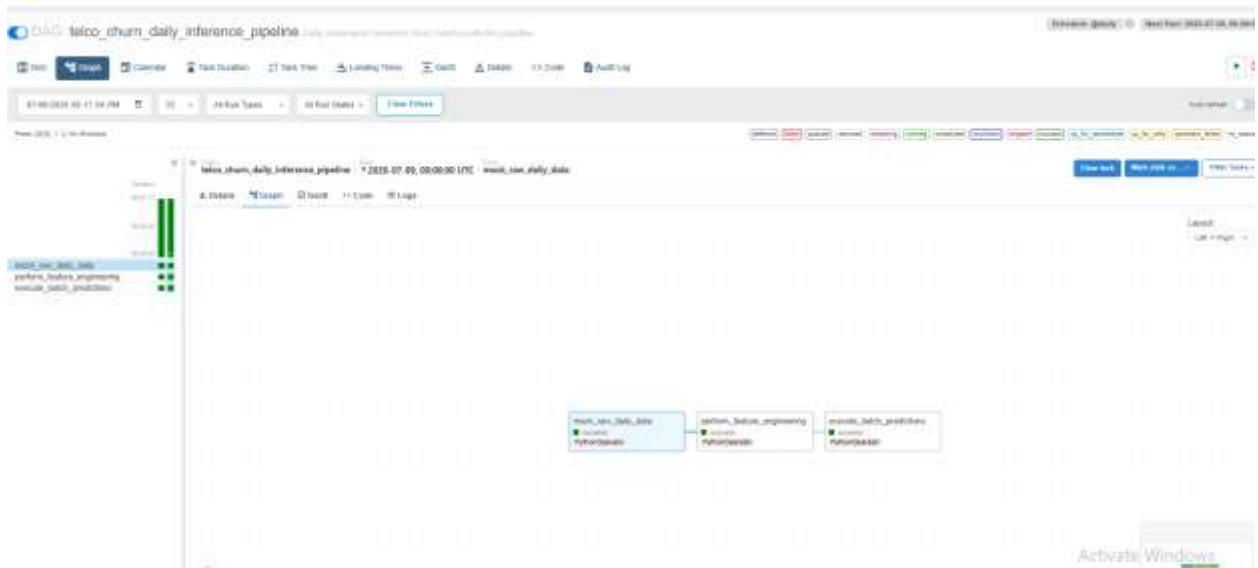




Project endpoints:

1. **Root Endpoint: GET /**
It serves the application's gui
2. **Health check endpoint: GET /health**
It verifies that the ML service is running correctly
3. **Prediction endpoint:**
It receives customer information and returns a churn prediction

→Airflow DAGs:





Automates the full daily batch churn prediction pipeline:

Task 1 → `mock_raw_data_task`:

Simulates pulling fresh customer records from a database.

FIX: previously returned the same 3 hardcoded rows every run. Now samples real, held-out customers from Milestone 1's `X_test.csv` (never used in training) and reverses the scaling/encoding Milestone 1 applied, so Task 2 can re-process them exactly like fresh incoming data. Which customers get sampled is seeded by the DAG's logical date, so a given date always reproduces the same batch (safe for retries/backfills) while different dates genuinely differ.

Task 2 → `feature_engineering_task`:

Replicates the exact feature engineering from Milestone 1:

- Encodes categorical fields (Contract, InternetService, PaymentMethod)
- Calculates `total_services` the SAME way as Milestone 1 SQL Query 8 (8 services: 6 add-ons + PhoneService + InternetService)

- Applies MinMaxScaler (loaded from Milestone 1 artifact) to the 3 continuous columns

Task 3 → batch_inference_task:

Loads best_churn_model.pkl (winner from Milestone 2 evaluation), runs predictions on the engineered batch, saves results.

FIX: previously overwrote a single fixed results file every run, so there was no history to plot drift against. Now keeps a dated snapshot per run AND a rolling history file. The history file is an idempotent upsert by run_date: re-running the same date replaces that date's rows only, so retries/backfills never create duplicates, and every other date's history is left untouched.

Flow: mock_data >> feature_engineering >> batch_predictions

Milestone 4 : Monitoring & Reporting

What it does:

- Logs every prediction automatically to a CSV file for drift monitoring
- Provides data for Power BI dashboard showing churn trends over time

Prediction Logging (automatic)

Every call to `/api/v1/churn/predict` automatically appends a row to:

milestone3-api-deployment/milestone3/app/data/predictions/prediction_logs.csv

Logged fields per prediction:

| Column | Description |

| `timestamp` | When the prediction was made |

| `tenure`, `MonthlyCharges`, `TotalCharges` | Core numeric features |

| `Contract`, `InternetService`, `PaymentMethod` | Key categorical features |

| `total\_services` | Number of subscribed services |

| `SeniorCitizen`, `Partner`, `Dependents` | Demographics |

| `PaperlessBilling`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`,
`TechSupport`, `StreamingTV`, `StreamingMovies` | Add-on services |

| `churn\_prediction` | 0 = stays, 1 = churns |

| `churn\_probability` | Model confidence (0.0–1.0) |

```
mpose.yml  README.md  logging.py  prediction_logs.csv M X  churn_inference_dag.py  run_pipeline.py  train_and_save_model.py 3
milestone3-api-deployment > milestone3 > app > data > predictions > prediction_logs.csv
You 28 minutes ago | 2 authors (Marim-1611 and one other)
1 timestamp,tenure,MonthlyCharges,TotalCharges,Contract,total_services,churn_prediction,churn_probability  #1:141, last word
2 2026-05-27 17:48:56.675683,12,75.0,980.0,1,4,0,0.25144529342651367
3 2026-05-29 18:12:13.320376,5,150.0,300.0,2,8,1,0.9141066670417786
4 2026-05-29 18:19:31.534128,24,120.0,3000.0,0,6,1,0.994277834892273
5 2026-05-29 18:28:52.816050,2,150.0,300.0,0,8,1,0.993077278137207
6 2026-05-29 18:29:04.052732,48,70.0,3500.0,2,4,0,0.12465681880712509
7 2026-05-29 18:29:11.241070,12,95.0,1200.0,1,5,1,0.8920326232910156
8 2026-05-29 18:29:20.739246,1,180.0,100.0,0,9,1,0.9959098100662231
9 2026-05-29 18:29:30.637336,8,130.0,950.0,0,7,1,0.9946630001068115
10 2026-05-29 18:29:38.286206,36,85.0,2800.0,1,5,1,0.7346135377883911
11 2026-05-29 18:29:47.231989,5,160.0,700.0,0,8,1,0.9923187494277954
12 2026-06-29 01:27:55.128731,2,95.0,190.0,Month-to-month,Fiber optic,Electronic check,1,1,0,0,1,0,0,0,0,0,1,0.8738165130565191
13 2026-06-29 01:27:55.177127,60,55.0,3300.0,Two year,DSL,Bank transfer (automatic),6,0,1,1,0,1,1,1,1,1,0,0.01141142999374403
14 2026-07-08 00:47:47.119369,12,70.05,840.6,One year,Fiber optic,Electronic check,3,0,1,0,1,1,0,1,1,0,0,0.23148152363123595 #2: 2
15 2026-07-08 00:48:45.048107,39,70.07,840.62,One year,Fiber optic,Electronic check,2,0,1,0,1,1,0,1,1,0,0,0.261692723611858156
16 2026-07-08 00:48:57.833862,39,70.07,840.62,One year,Fiber optic,Electronic check,2,0,1,0,1,1,0,1,1,0,0,0.2407643283871782
17 2026-07-08 00:49:02.933323,39,70.07,840.62,One year,Fiber optic,Electronic check,2,0,1,0,1,1,0,1,1,0,0,0.28985919845182956
18 2026-07-08 01:32:53.871113,39,70.07,840.62,One year,No,Electronic check,2,0,1,0,1,1,0,1,1,0,0,0.14924452456594725
19 2026-07-09 15:53:07.013405,12,70.05,840.6,One year,Fiber optic,Electronic check,3,0,1,0,1,1,0,1,1,0,0,0.23148152363123595 #3: 2
20 2026-07-09 18:20:46.398828,12,70.05,840.6,One year,Fiber optic,Electronic check,3,0,1,0,1,1,0,1,1,0,0,0.2691672520006147
21 2026-07-09 18:46:36.119983,12,70.05,840.6,One year,DSL,Electronic check,3,0,1,0,1,1,0,1,1,0,0,0.2249511042568245
22
```